

**The SPIDER Code - Axisymmetric Fixed Boundary Plasma Equilibrium Solver
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Вычислительный код SPIDER - расчет аксиально-симметричного равновесия плазмы с фиксированной границей

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Abstract

The SPIDER code is an axisymmetric fixed boundary plasma equilibrium solver for different formulations of the tokamak plasma equilibrium problem. Plasma with nested magnetic surfaces and a single magnetic axis limited by prescribed fixed boundary is assumed. Any reasonable set of two flux functions that can define toroidal current density profile (right hand side of the Grad-Shafranov equilibrium equation) can be prescribed as an input. High speed of the SPIDER code and high accuracy of computations make the code a powerful tool to solve complicated toroidal plasma equilibrium problems (such as plasma equilibrium with high beta, with high elongation of the plasma boundary, with low aspect ratio and with large Shafranov shift, with x-point at the plasma boundary, plasma profiles with reverse shear and with nonzero current density at plasma boundary etc.). Simple input and output formats make possible to use the SPIDER code in many tokamak plasma applications.

Аннотация

Данная работа представляет вычислительный код SPIDER, предназначенный для расчёта аксиально-симметричного равновесия плазмы с заданной фиксированной границей для различных формулировок задачи равновесия плазмы токамака. Предполагается, что плазма ограничена заданной фиксированной границей и обладает вложенными магнитными поверхностями при наличии единственной магнитной оси. В качестве входных параметров задачи могут быть заданы любые две из традиционно задаваемых потоковых функций, определяющих профиль плотности тороидального тока плазмы (правую часть уравнения равновесия Грэда-Шафранова). Высокая скорость и точность кода делают его мощным инструментом для решения сложных задач расчёта равновесия тороидальной плазмы, таких как: равновесие с высокими β , с большой вытянутостью границы плазмы, с малым аспектным отношением, с большим смещением магнитной оси, при задании границы плазмы с x-точкой, для профилей тороидальной плотности тока с обращенным широм, в случае ненулевой тороидальной плотности тока на границе плазмы и т.д. Простой и наглядный формат входных и выходных данных делает код SPIDER вполне доступным для эксплуатации и использования во многих приложениях, связанных с расчётами плазмы токамака.

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1. Introduction

Tokamak plasma equilibrium computation is a fundamental problem of magnetic confinement studies. Many plasma processes, including linear and early nonlinear stages of magneto-hydrodynamic (MHD) instabilities, plasma evolution and transport, plasma flows, waves and turbulence, represent different kinds of deviations from MHD equilibrium. Thus they require accurate calculations of plasma equilibrium configurations (see, for example, Refs. [1-8]).

In this paper we describe an axisymmetric fixed boundary plasma equilibrium solver - the SPIDER code - for solving the tokamak equilibrium problem based on the nonlinear Grad-Shafranov equilibrium equation with different sets of prescribed profiles, e.g. $dp/d\Psi$ and $F \cdot dF/d\Psi$, $dp/d\Psi$ and Q , $dp/d\Psi$ and $j_{\parallel}$. Here P is the plasma pressure, F is the poloidal current, Q is

the safety factor, $j_{\parallel}$ is the averaged parallel to the magnetic field component of the current density, Ψ is the poloidal flux. The SPIDER code output consists of the magnetic surfaces coordinates and other equilibrium magnetic field characteristics.

Because of the nonlinearity of the Grad-Shafranov equation all numerical methods for equilibrium calculation are iterative. Conventionally they can be subdivided into two classes:

- a.) *Eulerian*, that use a prescribed (e.g., rectangular or conformal to the plasma boundary) mesh calculation as, for example, in Refs. [8-14];
- b.) *Lagrangian*, that use curvilinear flux coordinates and adaptive to magnetic surfaces mesh for equilibrium calculations as in Refs. [7], [15-20], [26-35].

Eulerian methods have an advantage of being able to easily reproduce the two-dimensional geometry of complicated configurations as in Ref. [10]. They are widely used for simulations of equilibrium control in tokamaks as in Refs. [21], [36-38] and for the interpretation of experimental magnetic measurements (e.g. Refs. [22-24]). The disadvantage of those methods is in their limited range of formulations of equilibrium problems, restricted essentially to a prescribed right hand side of the Grad-Shafranov equation. Formulations requiring other input profiles are difficult to implement due to a necessity of frequent magnetic surface coordinate computing and averaging over them, as discussed in Ref. [25].

Lagrangian methods have an advantage in permitting to solve wide range of problems. Any set of two one-dimensional functions of the radial flux coordinate, that uniquely determines the current density and the pressure profile, can be used in flux coordinate methods. A disadvantage of flux coordinate methods consists in difficulties treating free boundary plasma equilibrium, especially with a separatrix (see, e.g., Ref. [7]).

Methods, that use flux coordinates, in turn, can be subdivided into two types:

- a.) variational (Refs. [16], [19], [28-30]) and inverse coordinate (Refs. [17], [31-32], [35]) methods;
- b.) adaptive grid methods (Refs. [15], [18], [33, 34]).

Variational and inverse coordinate codes solve equations in which unknowns are flux coordinates themselves. Adaptive grid codes solve equilibrium equation on given curvilinear grid and then use the numerical solution for advancing the computational grid and adjustment of the flux coordinate system.

The theory of perturbed equilibrium approach for solving the Grad-Shafranov equation for different formulations of the tokamak equilibrium problem has been described in Ref. [39]. The discrepancy between nonlinear equation and its linear analog in that work brings in the main contribution. Such approach is particularly adequate for the adaptive grid codes (such as Ref. [15]) and can use constraints, which are specific to the each equilibrium problem. However, it is more complicated to implement this approach in the variational methods or inverse variable codes due to the difficulty in the linearization of equilibrium equations used in those codes (although some attempts have been made, for example, in Ref. [16] with successful results presented in Ref. [35]).

In this paper we describe the SPIDER code - an axisymmetric fixed boundary plasma equilibrium solver, which is based on an adaptive grid approach.

The main restriction of the code consists in an assumption of nested magnetic surfaces with a single magnetic axis in plasma.

The main achievements of the code are in the increased computational speed and high accuracy of resolution of both differential plasma equilibrium problem and its discrete model.

The SPIDER code was thoroughly tested by means of both analytical tests and comparison with such well-known equilibrium solvers as POLAR-2D (KIAM, Ref. [32, 33]), CAXE (KIAM, Ref. [34]) and EFIT (GA, Ref. [40]).

At the present moment the SPIDER code is used in such ITER Central Team codes as PET (KIAM, Ref. [8, 38]), DINA (TRINITI, Ref. [36,37]) and ASTRA (Kurchatov Institute, Ref. [41]).

Despite of large number of tokamak fixed boundary plasma equilibrium codes and long history of their development, there is still a demand for accurate, fast and robust code, which can be used for extensive calculations of complicated toroidal plasma equilibrium problems: such as plasma equilibrium with high beta, high elongation of the plasma boundary, low aspect ratio and large Shafranov shift, with x-point at plasma boundary, plasma with reverse shear current density profile, and nonzero current density at plasma boundary etc. Large number of tests and real device numerical simulations show (in authors opinion) that the SPIDER code meets all the above-mentioned requirements.

2. Mathematical model

In the case of an axisymmetric plasma configuration in a magnetic plasma confinement device (tokamak type) the following vector equations:

$$\begin{aligned}\nabla p &= \vec{j} \times \vec{B}, \\ \mu_0 \vec{j} &= \nabla \times \vec{B}, \\ \nabla \cdot \vec{B} &= 0,\end{aligned}$$

(where p - gas kinetics isotropic plasma pressure, $\vec{B}$ - magnetic field vector, $\vec{j}$ - plasma current density vector) can be reduced into a well known scalar Grad-Shafranov equation given by (2.1), (2.2) for poloidal flux function ψ .

The SPIDER code is designed for numerical solving of the Grad- Shafranov axisymmetric fixed boundary plasma equilibrium problem

$$\Delta^* \psi(r, z) = -r \cdot \mu_0 \cdot j_\varphi(r, \psi), \quad (r, z) \in \Omega_b, \quad (2.1)$$

$$\Delta^* \psi = r^2 \nabla \cdot \left(\frac{1}{r^2} \nabla \psi \right) = r \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial \psi}{\partial r} \right) + \frac{\partial^2 \psi}{\partial z^2},$$

where (r, φ, z) - cylindrical coordinate system associated with plasma configuration symmetry axis, Ω - plasma domain in (r, z) plane, $\psi(r, z)$ is the poloidal magnetic flux function, $j_\varphi(r, \psi)$ is the toroidal plasma current density:

$$\mu_0 j_\varphi = r \mu_0 \frac{dp}{d\psi} + \frac{1}{r} f \frac{df}{d\psi}, \quad (2.2)$$

$$\vec{B} = \nabla \psi \times \nabla \varphi + f \nabla \varphi,$$

$$\vec{j} = \nabla f \times \nabla \varphi + r j_\varphi \nabla \varphi,$$

$$\psi = r A_\varphi = \Psi / 2\pi, \quad f = \mu_0 r B_\varphi = \mu_0 F / 2\pi.$$

Here $p(\psi)$ is the plasma pressure profile, A_φ is the toroidal component of the magnetic field vector potential, B_φ is the toroidal component of the magnetic field, Ψ is the poloidal magnetic flux (measured from symmetry axis $r = 0$), F is the poloidal current (measured from the symmetry axis $r = 0$):

$$\Psi(r, z) = \iint_{S_P} (\vec{B} \cdot \vec{n}) dS, \quad F(r, z) = \iint_{S_P} (\vec{j} \cdot \vec{n}) dS,$$

where S_P is the poloidal surface, enclosed by circular line in toroidal direction passing through the (r, z) point.

Assuming presence of the nested surfaces with a single magnetic axis we can define plasma toroidal flux and current in the following manner:

$$\Phi(\psi) = \iint_{S_t(\psi)} (\vec{B} \cdot \vec{n}) dS = \iint_{S_t(\psi)} f(\psi) \frac{drdz}{r},$$

$$J(\psi) = \iint_{S_t(\psi)} (\vec{j} \cdot \vec{n}) dS = \iint_{S_t(\psi)} j_\varphi(r, \psi) drdz,$$

where $S_t(\psi)$ is toroidal cross-section ($\varphi = const$) of magnetic surface $\psi(r, z) = const$.

The following boundary condition at the plasma boundary curve $\partial\Omega_b$ is used in the SPIDER code:

$$\psi(r, z) = \psi_{bound} = 0, \quad (r, z) \in \partial\Omega_b. \quad (2.3)$$

It is assumed that plasma boundary $\partial\Omega_b$ is prescribed. There are two possibilities to set plasma boundary $\partial\Omega_b$ in the SPIDER code:

1. by means of the following analytical formulas:

$$\begin{aligned}
 r_b &= r_0 + \frac{r_0}{A} \cdot \cos(\Theta + \Delta \sin \Theta) \\
 z_b &= z_0 + \frac{r_0}{A} k \cdot \sin \Theta
 \end{aligned}
 \tag{2.4}$$

where (r_0, z_0) are coordinates of the geometric center of the plasma boundary, Θ - poloidal angle in (r, z) plane, A is the plasma boundary aspect ratio, k is the plasma boundary elongation and Δ is the plasma boundary triangularity:

$$\begin{aligned}
 \Delta &= 0.5 \cdot [\Delta_u + \Delta_l + (\Delta_u - \Delta_l) \cdot \sin \Theta] \\
 k &= 0.5 \cdot [k_u + k_l + (k_u - k_l) \cdot \sin \Theta]
 \end{aligned}
 \tag{2.5}$$

where k_u, k_l are the upper and lower elongations respectively, Δ_u, Δ_l are the upper and lower triangularities respectively. Of course it is possible to use any other appropriate analytical formulas.

2. by means of prescribed boundary points coordinates array: $r_b(i), z_b(i), i = 1, \dots, N_b$.

There are following settings for axisymmetric fixed boundary equilibrium problem in the SPIDER code:

1. Setting with **prescribed** $j_\varphi(r, z)$ - **right hand side** (2.2) of the Grad-Shafranov equation (2.1) and toroidal plasma current value I_{pl} i.e. with prescribed profiles $p'_\psi(\bar{\psi}), f f'_\psi(\bar{\psi})$ as functions of normalized poloidal flux $\bar{\psi} = \frac{\psi - \psi_{axis}}{\psi_{bound} - \psi_{axis}} \in [0, 1]$ and with toroidal plasma current value given by

$$I_{pl} = \iint_{\Omega_b} j_\varphi dr dz
 \tag{2.6}$$

There are two possibilities to set profiles $p'_\psi(\bar{\psi}), ff'_\psi(\bar{\psi})$ in the SPIDER code:

- a.) by means of the analytical formulas:

$$\begin{aligned} p'_\psi &= \alpha_0 (1 - \bar{\psi}^{\alpha_1})^{\alpha_1}, \\ ff'_\psi &= \beta_0 (1 - \bar{\psi}^{\beta_1})^{\beta_1}, \end{aligned} \quad (2.7)$$

Of course it is possible to use any other appropriate analytical formulas.

- b.) by means of prescribed arrays $\bar{\psi}_i, p'_\psi(\bar{\psi}_i), ff'_\psi(\bar{\psi}_i), i = 1, \dots, N_{pf}$.

2. Setting with **prescribed profiles** $p'_\psi(\bar{\psi})$ and **safety factor** $q(\bar{\psi}) = -\frac{d\Phi}{d\psi}$ as functions of normalized poloidal flux $\bar{\psi} = \frac{\psi - \psi_{axis}}{\psi_{bound} - \psi_{axis}} \in [0, 1]$ and with prescribed value of poloidal flux in plasma $\psi_{axis} - \psi_{bound}$. Inside plasma domain the poloidal current function $f(\psi)$ is connected with safety factor $q(\psi)$ as follows:

$$f(\psi) = \frac{2\pi q(\psi)}{\oint_{\psi=const} \frac{dl}{r|\nabla\psi|}},$$

where integral is taken along the magnetic surface $\psi(r, z) = const$ toroidal cross-section line.

There are two possibilities to set profile $p'_\psi(\bar{\psi})$ for the problem setting 2 with the prescribed safety factor profile:

- a.) by means of analytical formula:

$$p'_\psi = \alpha_0 (1 - \bar{\psi}^{\alpha_1})^{\alpha_1};$$

Of course it is possible to use any other appropriate analytical formulas.

b.) by means of prescribed arrays: $\bar{\psi}_i, p'_\psi(\bar{\psi}_i), i = 1, \dots, N_p$.

There is another possibility to set the profile of $q(\bar{\psi})$ in the SPIDER code by means of prescribed arrays: $\bar{\psi}_i, q(\bar{\psi}_i), i = 1, \dots, N_q$.

In any considered setting of the equilibrium problem there is possibility to prescribe the value of poloidal beta β_p by means of rescaling the pressure profile $p'_\psi(\bar{\psi})$.

In the SPIDER code the value of poloidal beta β_p is defined in the following manner:

$$\beta_p = \frac{8\pi \int_{\Omega_b} p dS}{\mu_0 I_{pl}^2}.$$

The total beta β is defined as:

$$\beta = \frac{2\langle p \rangle_V}{B_0^2},$$

where B_0 - vacuum toroidal field value at the plasma center and

$$\langle p \rangle_V = \frac{\int_{V_b} p dV}{\int_{V_b} dV},$$

where V_b - the plasma volume.

3. Finite-difference scheme

The computational domain Ω_b is covered by a computational grid, which is topologically equivalent to a radially-annular grid and it is used as initial guess for construction of final magnetic surface adaptive grid.

The SPIDER code discrete model is based on magnetic surface adaptive grid finite-difference method with 9-points difference scheme.

A priori unknown adaptive grid has radially-annular structure:

$$\{z_{\psi}(a_i, \theta_j), z_{\bar{\psi}}(a_i, \theta_j)\}, \quad i=1, \dots, N_a, j=1, \dots, N_{\theta}, \quad (3.1)$$

where $a \equiv a(\psi)$ is the adaptive radial direction variable, θ is the poloidal angle direction variable. Grid lines $a_i = a(\psi_i) = \text{const}$ form the set of nested magnetic surfaces where $a_1 = a(\psi = \psi_{\text{axis}})$ labels magnetic axis location, $a_i = a(\psi_i)$ labels magnetic surface $\psi = \psi_i = \text{const}$ and $a_{N_a} = a(\psi = \psi_{\text{bound}})$ labels plasma boundary curve $\partial\Omega_b$. Grid lines $\theta = \text{const}$ are chosen to be straight.

The difference analog of the Grad-Shafranov operator $\Delta^* \psi$ (2.1) is constructed on the basis of the conservative finite-difference approximation of the $\nabla \times \nabla \times$ operator by means of the operator-variational scheme [42].

Let us consider a vector potential $\vec{A}$ of the poloidal magnetic field $\vec{B}_p = \nabla \psi \times \nabla \varphi$ with only one non-trivial toroidal projection, connected with the poloidal magnetic flux $\psi(r, z)$ as follows:

$$\vec{A} = A_{\varphi} \vec{e}_{\varphi} = A_{\varphi} r \nabla \varphi = \psi \nabla \varphi, \quad (3.2)$$

where $\vec{e}_{\varphi} = r \nabla \varphi = \nabla \varphi / |\nabla \varphi|$ - the unit vector in toroidal angle φ direction. Then Grad-Shafranov operator $\Delta^* \psi$ (2.1) can be written in the form of the non-trivial projection of the $\nabla \times \nabla \times \vec{A}$ operator

on toroidal angle φ direction:

$$(-\Delta^* \psi) \nabla \varphi = \nabla \times \nabla \times \vec{A}, \quad (3.3)$$

or in a scalar form

$$-\Delta^* \psi = r (\nabla \times \nabla \times \vec{A})_{\varphi}. \quad (3.4)$$

For any two dimensional cell of our 2D radially-annular grid we can construct corresponding three dimensional cell, which is a body of rotation of this 2D cell in the φ direction.

The difference approximation of the vector potential $\vec{A}$ is determined by means of the orthogonal projections on the edges of the 3D cell. In our case we have only non-trivial projections on the edges in toroidal direction:

$$A_{\varphi}^{ij} = \frac{\psi_{ij}}{r_{ij}}. \quad (3.5)$$

The difference approximation of the $\nabla \times \vec{A}$ operator in terms of its orthogonal projections on the unit normals to the faces of the 3D cell is constructed on the basis of the invariant definition, that is valid for an arbitrary infinitesimal area S with a boundary curve ∂S :

$$(\nabla \times \vec{A}) \cdot \vec{n} = \lim_{S \rightarrow 0} \frac{\oint_{\partial S} \vec{A} \cdot d\vec{l}}{S}, \quad (3.6)$$

where $\vec{n}$ is the unit vector normal to the S face. A direct approximation of (3.6) on the faces of the 3D

cell gives the following approximation of the orthogonal projections of the $\nabla \times \vec{A}$:

$$(\nabla \times A)_{i+1/2,j,k+1/2} = \frac{A_{\varphi}^{i+1j} \cdot r_{i+1j} - A_{\varphi}^{ij} \cdot r_{ij}}{0.5 \cdot l_{i+1/2j} (r_{i+1j} + r_{ij})} = \frac{\psi_{i+1j} - \psi_{ij}}{0.5 \cdot l_{i+1/2j} (r_{i+1j} + r_{ij})}, \quad (3.7)$$

$$(\nabla \times A)_{i+1/2,j+1,k+1/2} = \frac{A_{\varphi}^{i+1j+1} \cdot r_{i+1j+1} - A_{\varphi}^{ij+1} \cdot r_{ij+1}}{0.5 \cdot l_{i+1/2j+1} (r_{i+1j+1} + r_{ij+1})} = \frac{\psi_{i+1j+1} - \psi_{ij+1}}{0.5 \cdot l_{i+1/2j+1} (r_{i+1j+1} + r_{ij+1})},$$

$$(\nabla \times A)_{i,j+1/2,k+1/2} = -\frac{A_{\varphi}^{ij+1} \cdot r_{ij+1} - A_{\varphi}^{ij} \cdot r_{ij}}{0.5 \cdot l_{ij+1/2} (r_{ij+1} + r_{ij})} = -\frac{\psi_{ij+1} - \psi_{ij}}{0.5 \cdot l_{ij+1/2} (r_{ij+1} + r_{ij})},$$

$$\begin{aligned} (\nabla \times A)_{i+1,j+1/2,k+1/2} &= -\frac{A_{\varphi}^{i+1j+1} \cdot r_{i+1j+1} - A_{\varphi}^{i+1j} \cdot r_{i+1j}}{0.5 \cdot l_{i+1j+1/2} (r_{i+1j+1} + r_{i+1j})} \\ &= -\frac{\psi_{i+1j+1} - \psi_{i+1j}}{0.5 \cdot l_{i+1j+1/2} (r_{i+1j+1} + r_{i+1j})}, \end{aligned}$$

where $l_{i+1/2j}$, $l_{i+1/2j+1}$, $l_{ij+1/2}$ and $l_{i+1j+1/2}$ are corresponding lengths of the 2D cell edges. These projections are related to the centers of the corresponding faces of the 3D cell.

In turn the difference approximation of the $\nabla \times \nabla \times \vec{A}$ operator is determined by means of the orthogonal projections on the edges of the 3D cell and is related to the their centers. For construction of

the operator the difference analog of the following integral theorem is used:

$$\int_V (b \cdot \nabla \times \nabla \times a) dV = \int_{\partial V} (\nabla \times a) \times b \cdot dl + \int_V (\nabla \times a) \cdot (\nabla \times b) dV \quad (3.8)$$

4. Iteration process

Let us denote finite difference approximation of the Grad-Shafranov equation (2.1) on the adaptive grid (3.1) by means of subscript h

$$L_h \psi_h \equiv \Delta_h^* \psi_h = -r_h \cdot \mu_0 \cdot j_{\varphi_h}(r_h, \psi_h) \quad (4.1)$$

The final purpose of solving of the discrete problem (4.1) with corresponding boundary conditions approximation consists in the finding both unknown coordinates of magnetic surface adaptive grid (3.1) and desired unknown discrete function ψ_h in the nodes of this grid.

Let us assume that the adaptive grid $\{r_{\psi}^s(a_i, \theta_j), z_{\psi}^s(a_i, \theta_j)\}$ coordinates and the corresponding numerical solution $\psi_h^s \equiv \psi_i^s$ for iteration number s are known. Then the iteration loop of the SPIDER code consist of the following two parts.

The first part implies that we solve the linear problem with respect to ψ_h^{s+1} on known adaptive grid $\{r_{\psi}^s(a_i, \theta_j), z_{\psi}^s(a_i, \theta_j)\}$:

$$L_h^s \psi_h^{s+1} \equiv \Delta_h^{*s} \psi_h^{s+1} = -r_h^s \cdot \mu_0 \cdot j_{\varphi_h}(r_h^s, \psi_h^s) \quad (4.2)$$

The second part implies that we construct new $(s+1)$ approximation of the unknown adaptive

grid $\{r_{\psi}^{s+1}(a_i, \theta_j), z_{\psi}^{s+1}(a_i, \theta_j)\}$ assuming that prescribed $\alpha_i = \text{const}$ grid lines coincide with $\psi_h^{s+1}(a_i) = \text{const}$ lines of closed magnetic surfaces.

Let us note that the matrix L_h^s of the linear algebraic equations (4.2) depends on adaptive grid coordinates. Therefore, we need to invert this matrix on each iteration step $\mathcal{S}$. As the matrix inversion on each iteration step is very expensive, a combination of direct and iterative methods is employed to solve the equations (4.2) more efficiently in course of adaptive grid iterations. With this in mind let us write

$$L_h^{s+k} = \left(L_h^s + \delta L_h^{s+k} \right), \quad k \geq 1. \quad (4.3)$$

Because matrix L_h^s changes very slowly from iteration to iteration, the inverse matrix $\left(L_h^s \right)^{-1}$ is close to the matrix $\left(L_h^{s+k} \right)^{-1}$. Therefore we can write

$$\left(L_h^{s+k} \right)^{-1} = \left(L_h^s + \delta L_h^{s+k} \right)^{-1} \approx \left(L_h^s \right)^{-1} - \left(L_h^s \right)^{-1} \cdot \delta L_h^{s+k} \cdot \left(L_h^s \right)^{-1}, \quad (4.4)$$

where matrix $\left(L_h^{s+k} \right)^{-1}$ can be used as a "preconditioner" for some fast iterative method.

Finally instead of the matrix inversion on each iteration step we use the following iterative procedure:

- a.) for iteration step $\mathcal{S}$ we compute the inverse matrix $\left(L_h^s \right)^{-1}$ and solve equations (4.2);
- b.) for the next k iteration steps equations (4.2) are solved by means of some fast iterative method using as a "preconditioner" the matrix in the right hand side of the equation (4.4):

$$C_h^{s+k} = \left(L_h^s \right)^{-1} - \left(L_h^s \right)^{-1} \cdot \delta L_h^{s+k} \cdot \left(L_h^s \right)^{-1}.$$

In the SPIDER code the following variants of the grid adaptation variable $a \equiv a(\psi)$ may be used:

$$1. \quad a = \bar{\psi},$$

$$2. \quad a = \sqrt{\bar{\psi}},$$

$$3. \quad a = \sqrt{\bar{\Phi}},$$

where $\bar{\psi} = \frac{\psi - \psi_{axis}}{\psi_{bound} - \psi_{axis}} \in [0,1]$ - normalized poloidal flux, $\bar{\Phi} = \frac{\Phi}{\Phi_{bound}} \in [0,1]$ - normalized toroidal flux.

Let us consider in more detail iteration process for equilibrium problem with prescribed safety factor $q(\bar{\psi})$ profile.

On the basis of averaged 1D Kruskal-Kulsrud equilibrium equation

$$J' \cdot \Psi' = F' \cdot \Phi' + p' \cdot V' \quad (4.5)$$

and relations between currents and fluxes in axisymmetric configuration

$$J = \alpha_{22} \cdot \Psi', \quad F = \alpha_{33} \cdot \Phi', \quad q = -\frac{\Phi'}{\Psi'}, \quad F = -\alpha_{33} \cdot q \cdot \Psi', \quad (4.6)$$

we can rewrite averaged 1D equilibrium equation in the following form

$$(\alpha_{22} \cdot \Psi')' = q \cdot (\alpha_{33} \cdot q \cdot \Psi')' + \frac{dp}{d\psi} \cdot V'. \quad (4.7)$$

Here coefficients α_{22}, α_{33} are determined only by means of magnetic surfaces geometry. Assuming

that the derivative $\left(\frac{\bullet}{V}\right)' = \frac{d(\bullet)}{dV}$ in (4.6) , we define the coefficients α_{22}, α_{33} with the help of the following formulae

$$\alpha_{22} = \left\langle \frac{|\nabla V|^2}{r^2} \right\rangle_V, \quad \alpha_{33} = \left\langle \frac{1}{r^2} \right\rangle_V, \quad (4.8)$$

where V is the magnetic surface volume.

On the basis of 1D relation (4.7) the following iteration process works very well for solving fixed boundary tokamak equilibrium problem with prescribed safety factor $q(\psi)$ profile:

- (1.) let us know magnetic surface geometry and all required magnetic surface functions $\psi^n, \alpha_{22}^n, \alpha_{33}^n, V^n$ on iteration number n ;
- (2.) we can solve 1D equation (4.7) on "frozen" 1D flux grid

$$\frac{d}{dV} \left(\alpha_{22}^n \cdot \frac{d\tilde{\psi}}{dV} \right) - q \cdot \frac{d}{dV} \left(\alpha_{33}^n \cdot q \cdot \frac{d\tilde{\psi}}{dV} \right) = \frac{dp}{d\psi}$$

with prescribed boundary values ψ_{axis} and ψ_{bound} .

$$\text{and obtain } \tilde{\psi}^{n+1} \text{ and } \left(F \frac{dF}{d\psi} \right)^{n+1} = \alpha_{33}^n \cdot q \cdot \frac{d}{dV} \left(\alpha_{33}^n \cdot q \cdot \frac{d\tilde{\psi}^{n+1}}{dV} \right) ;$$

- (3.) we can make the Picard iteration for solving of the 2D Grad-Shafranov equilibrium equation

(2.1) on "frozen" 2D grid (3.1)

$$\frac{1}{r} \Delta^* \psi^{n+1} = -\frac{1}{r} \left(F \frac{dF}{d\psi} \right)^{n+1} - \mu_0 \cdot r \cdot \frac{dp}{d\psi}$$

and on the basis of this solution fulfill the construction of new approximation of 2D adaptive grid (i.e. magnetic surfaces coordinates);

- (4.) after step (3.) we can compute all required magnetic surface functions $\psi^{n+1}, \alpha_{22}^{n+1}, \alpha_{33}^{n+1}, V^{n+1}$ on iteration number $(n+1)$ and go to the beginning of the iteration loop.

Let us consider in more detail iteration process for equilibrium problem with prescribed $\langle \vec{j} \cdot \vec{B} \rangle_V$ profile.

Taking into account that

$$\langle \vec{j} \cdot \vec{B} \rangle_V \cdot V' = \mu_0 (F \cdot J' - J \cdot F') \quad (4.9)$$

and relations (4.5), (4.6) we can obtain the following set of two 1D nonlinear equations for unknowns Ψ and Φ :

$$(\alpha_{22} \Psi')' \cdot \Psi' = (\alpha_{33} \Phi')' \cdot \Phi' + p' V' \quad , \quad (4.10)$$

$$\mu_0 \left[\alpha_{33} \Phi' \cdot (\alpha_{22} \Psi')' - \alpha_{22} \Psi' \cdot (\alpha_{33} \Phi')' \right] = \left\langle \vec{j} \cdot \vec{B} \right\rangle_V \cdot V'$$

with the following boundary conditions:

$$\Phi_{axis} = 0 \text{ and } \alpha_{33} \cdot \frac{d\Phi}{dV} = F_{bound},$$

where F_{bound} - vacuum poloidal current value – is prescribed.

We can solve system of 1D equations (4.10) on "frozen" 1D flux grid to obtain $\tilde{\psi}^{n+1}$, $\tilde{\Phi}^{n+1}$ and taking into account (4.6) relations between currents and fluxes, and get $\left(F \frac{dF}{d\psi} \right)^{n+1}$.

As in the case of prescribed safety factor $q(\bar{\psi})$ profile, we can make the Picard iteration for solving of the 2D Grad-Shafranov equilibrium equation (2.1) on "frozen" 2D grid (3.1)

$$\frac{1}{r} \Delta^* \psi^{n+1} = -\frac{1}{r} \left(F \frac{dF}{d\psi} \right)^{n+1} - \mu_0 \cdot r \cdot \frac{dp}{d\psi},$$

and on the basis of this solution fulfill the construction of new approximation of 2D adaptive grid (i.e. magnetic surfaces coordinates).

5. Test simulations

Validation of the code against exact solution on the set of the different grid sizes and plasma

parameters has been carried out. For this test the following formulas of the exact solution were taken from [43], p.132 :

$$\psi(r, z) = c_0 \left[(\varepsilon^2 - r^2)(r^2 - \delta^2) - 4\alpha^2(r^2 - \sigma^2)z^2 \right], \text{ where } \alpha > 0, 0 \leq \sigma \leq \delta < \varepsilon,$$

$$\frac{dp}{d\psi} = 8(1 + \alpha^2)c_0, \quad \frac{dF^2}{d\psi} = -16\alpha^2\sigma^2c_0,$$

$$\psi_{axis} = \frac{c_0}{4}(\varepsilon^2 - \delta^2)^2, \quad r_{axis}^2 = \frac{\varepsilon^2 + \delta^2}{2}, \quad z_{axis} = 0.$$

Parameter $C_0 = \frac{4}{(\varepsilon^2 - \delta^2)^2}$ was chosen to satisfy $\psi_{axis} = 1$.

The results of simulations for different values of plasma aspect ratio A , elongation k and triangularity Δ are presented at the table № 1. Calculated values of coordinate of magnetic axis R_{axis}^C and corresponding poloidal flux value Ψ_{axis}^C are compared to the exact values $R_{axis}, \psi_{axis} = 1$ for different values of grid size $N_a \times N_\theta$. It is seen that difference between numerical and exact solution $\|\varepsilon_\psi\|_C$ is quite small. It is necessary to point out, that accuracy of numerical solution convergence from grid size is quadratic.

Table № 1.

A	k	Δ	R_{axis}	R_{axis}^C	Ψ_{axis}^C	$N_a \times N_\theta$	$\ \varepsilon_\psi\ _C$
1.03	4.	0.78	0.7072	0.7075	0.998	64*127	$3 \cdot 10^{-3}$

5.	2.	0.1	1.2748	1.27472	0.998	32*64	$2.5 \cdot 10^{-3}$
				1.27474	0.9995	64*127	$6.2 \cdot 10^{-4}$
2.	2.	0.268	1.118	1.1179	0.998	32*64	$2.5 \cdot 10^{-3}$
				1.11799	0.9995	64*127	$6.2 \cdot 10^{-4}$
2.	8.22	0.3	1.118	1.1179	0.99998	32*64	$9. \cdot 10^{-3}$
				1.11800	$1. \cdot 10^{-6}$	64*127	$1. \cdot 10^{-3}$
				1.11800	$1. \cdot 2. \cdot 10^{-7}$	128*255	$3.1 \cdot 10^{-4}$

6. Examples of runs and Figures

Several examples of the SPIDER code runs are presented in this section.

The first case corresponds to high beta poloidal $\beta_p \approx 4.5$ and prescribed reverse shear averaged current density profile $\langle J_\varphi \rangle$ with circular plasma boundary shape. Results of simulation are shown in Figs. 1-3. It is seen that magnetic axes is shifted far away from the plasma geometrical center.

In the second case, plasma equilibrium with a high plasma boundary elongation $k = 6$ is calculated. This case demonstrates that code can reliably calculate quite exotic equilibrium configuration without problems. Results of simulation are shown in Figs. 4-6.

As an example for the third case, plasma boundary with a single X-point has been chosen. This example shows, that code can be used to simulate realistic tokamak experiments with diverted plasma. Results are shown in the Figs. 7-10.

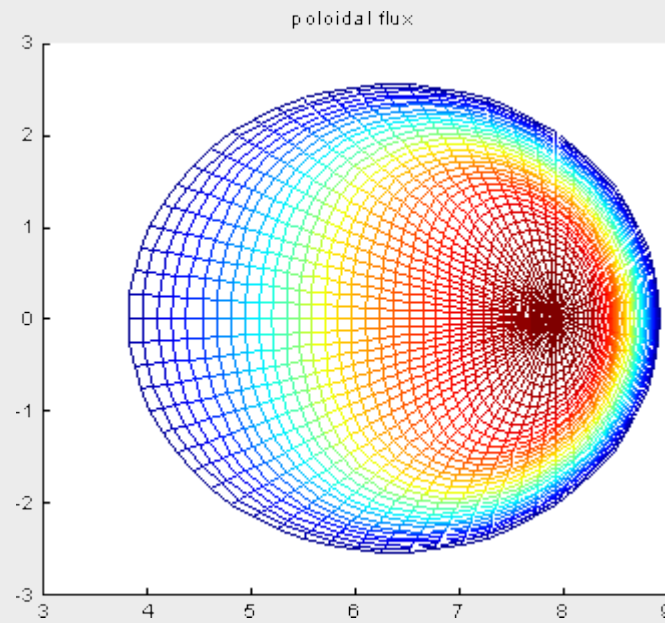


Fig.1 Magnetic surfaces adaptive grid: $\beta_p \approx 4.5$, given $\langle j_\varphi \rangle$.

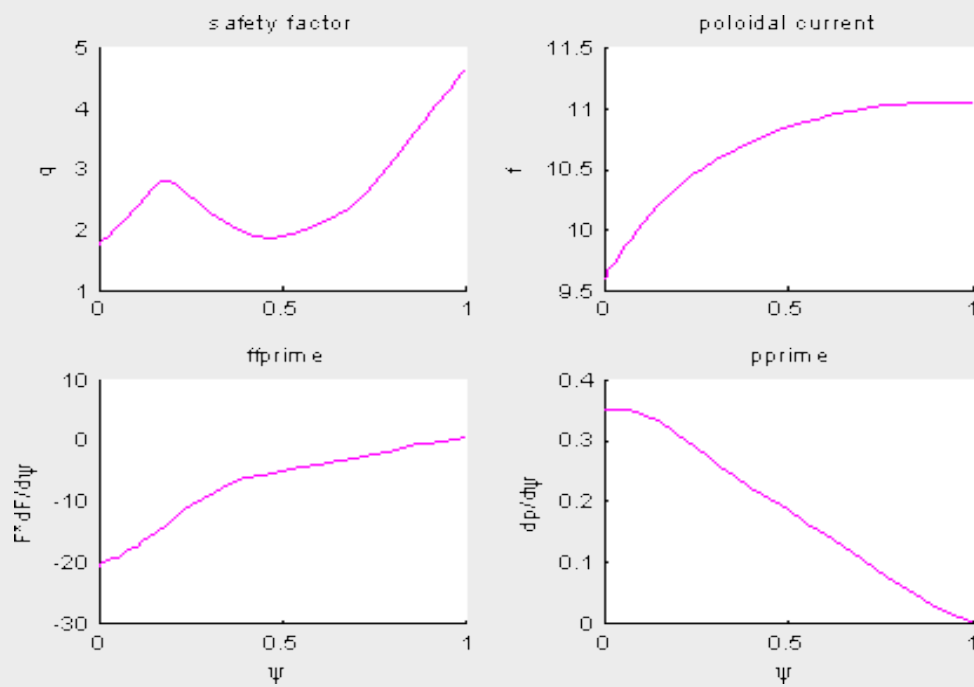


Fig.2 Plasma profiles versus normalized ψ : safety factor q , poloidal current I , current density profile parameters $F \cdot \frac{dF}{d\psi}$ and $\frac{dp}{d\psi}$

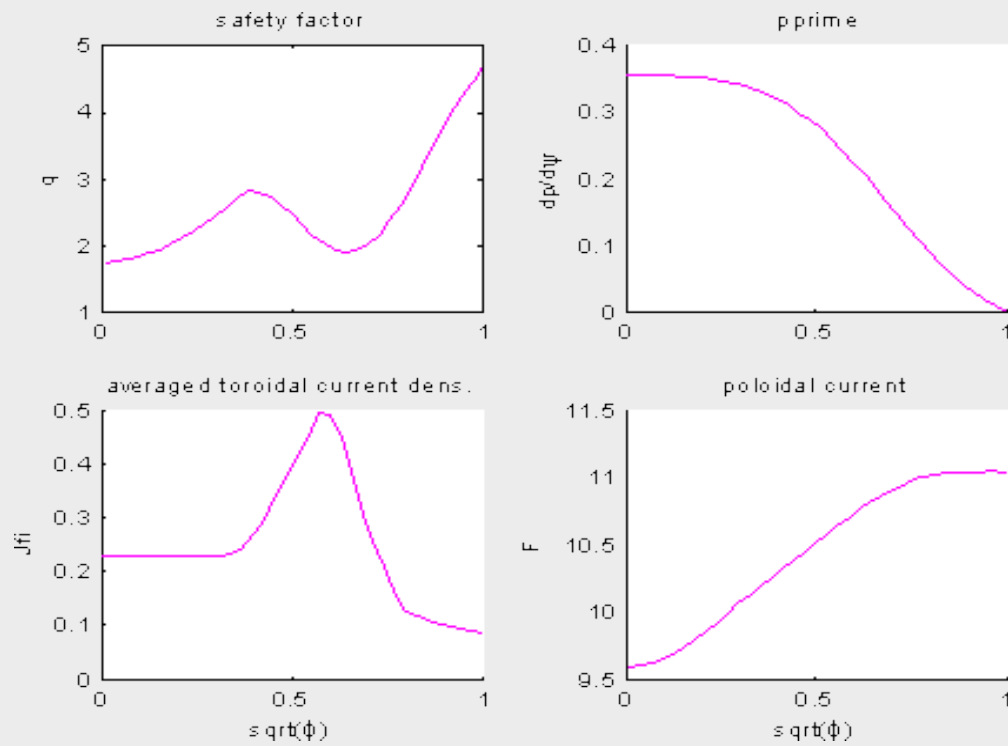


Fig.3. Plasma profiles versus normalized $\sqrt{\Phi}$: safety factor q , poloidal current F , current density profile parameters $\frac{dp}{d\psi}$ and averaged $\langle j_\phi \rangle$.

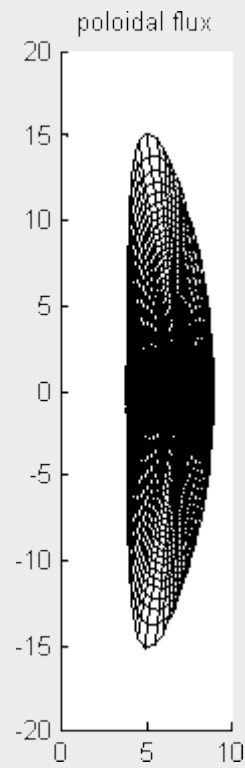


Fig.4. Magnetic surfaces adaptive grid: plasma elongation $k = 6$.

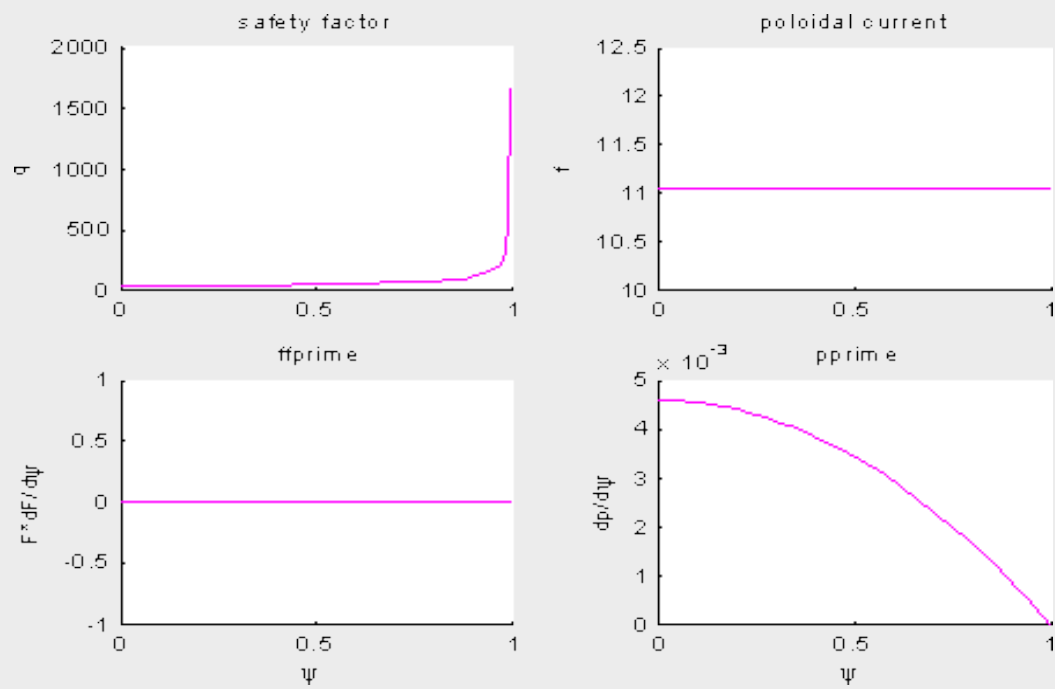


Fig.5. Plasma profiles versus normalized ψ : safety factor q , poloidal current j , current density

profile parameters $F \cdot \frac{dF}{d\psi}$ and $\frac{dp}{d\psi}$.

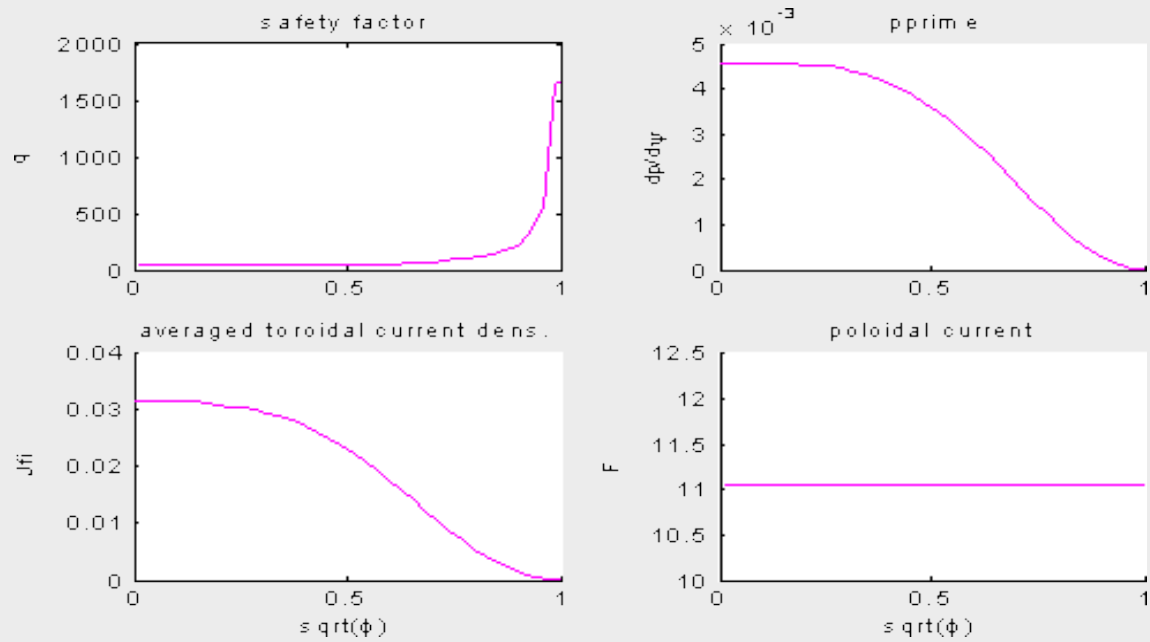


Fig.6. Plasma profiles versus normalized $\sqrt{\Phi}$: safety factor Q , poloidal current F , current density profile parameters $\frac{dp}{d\psi}$ and averaged $\langle J_{\phi} \rangle$.

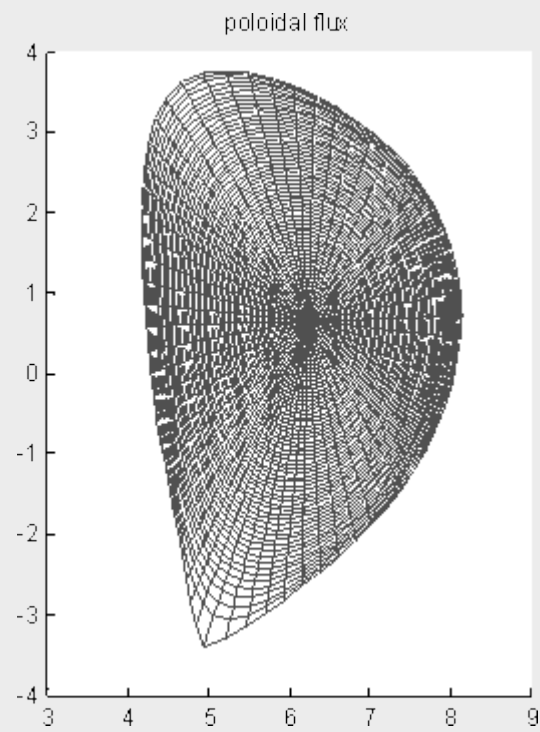


Fig.7. Magnetic surfaces adaptive grid: diverted plasma.

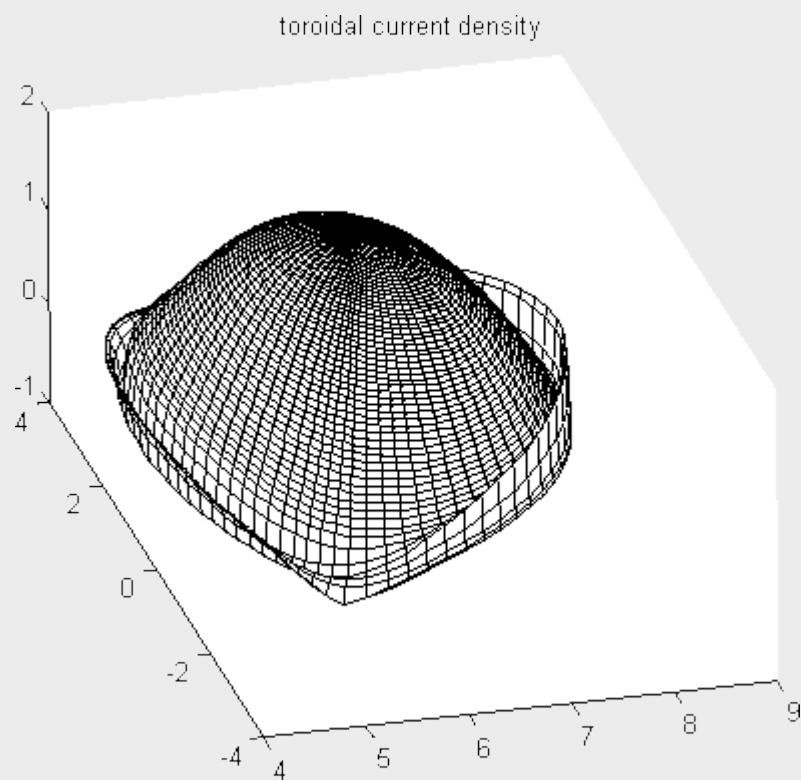


Fig.8. Plasma toroidal current density: diverted plasma.

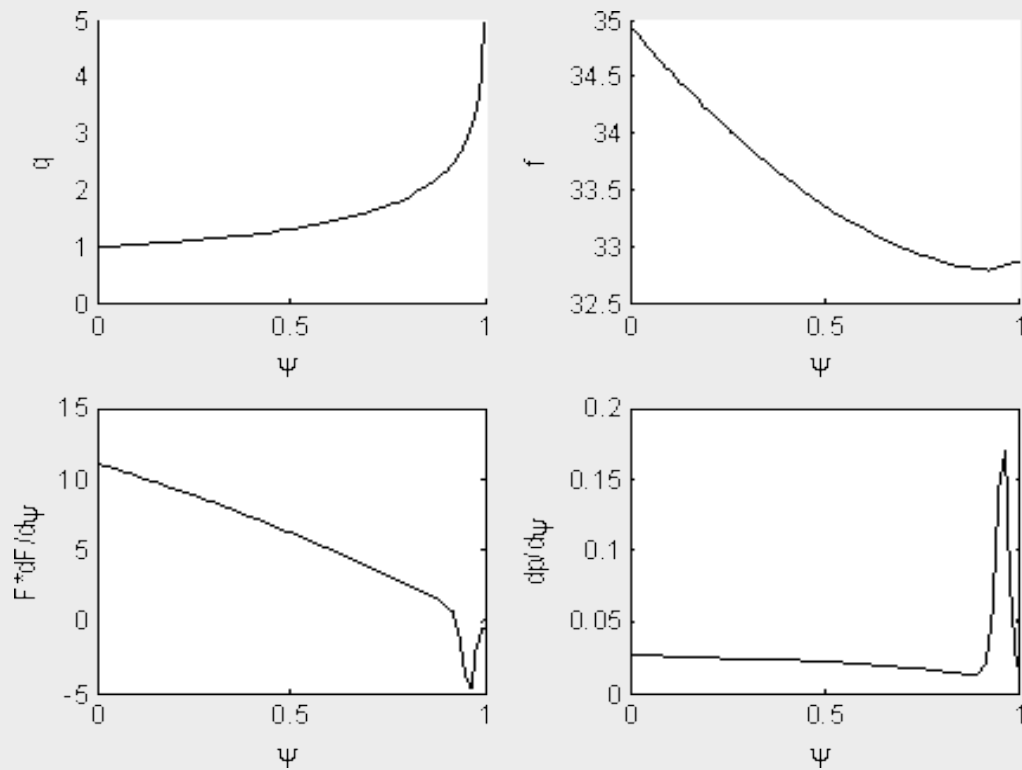


Fig.9. Plasma profiles versus normalized ψ : safety factor q , poloidal current F , current density $F \cdot \frac{dF}{d\psi}$ and $\frac{dp}{d\psi}$.

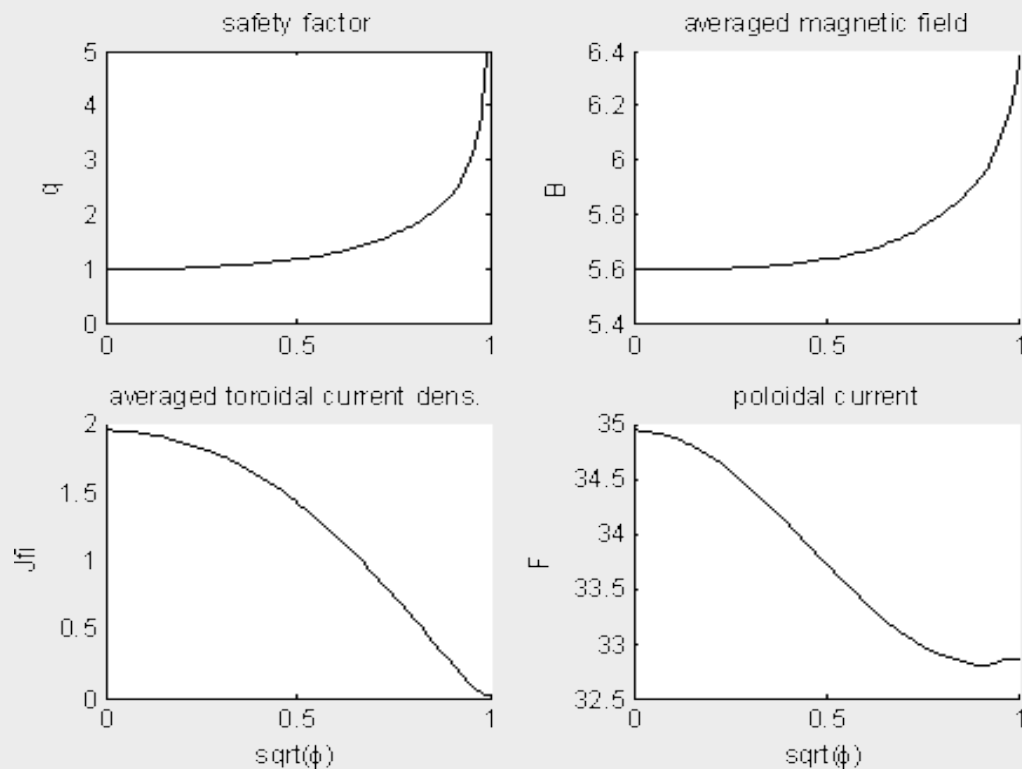


Fig.10. Plasma profiles versus normalized $\sqrt{\Phi}$: safety factor Q , averaged magnetic field $\langle B \rangle$, averaged toroidal current density $\langle j_\varphi \rangle$, poloidal current F' .

7. References

- [1] V.D.Shafranov, *Reviews of Plasma Physics* (Consultants Bureau, New York, 1966), Vol. 2, p. 103.
- [2] H.Grad and J.Hogan, *Phys. Rev. Lett.* **24**, 1337 (1970).
- [3] H.Grad, P.N.Hu and D.C.Stevens, *Proc. Natl. Acad. Sci. USA* **72**, 3789 (1975).
- [4] L.E.Zakharov and V.D.Shafranov, *Reviews of Plasma Physics* (Consultants Bureau, New York, 1981), Vol. 11, p. 153.
- [5] R.R.Khairutdinov and V.E.Lukash, *J. Comput. Phys.* **109**, 193 (1993).
- [6] L.Degtyarev, A.Martynov, S.Medevedev, F.Troyon, L.Vilard and R.Gruber, *Comput. Phys. Commun.* **103**, 10 (1997).
- [7] S.A.Galkin, V.V.Drozdov, A.A.Ivanov, S.Yu.Medvedev and Yu.Yu.Poshekhonov, *General Atomic Report GA-A23045*, May 1999.
- [8] S.A.Galkin, A.A.Ivanov, S.Yu.Medvedev and Yu.Yu.Poshekhonov, *Nucl. Fusion* **37**, No. 10, 1455 (1997).
- [9] W.Feneberg and K.Lackner, *Nucl. Fusion* **13**, 549 (1973).
- [10] F.J.Helton and T.S.Wang, *Nucl. Fusion* **18**, 1523 (1978).
- [11] J.L.Lohnson, H.E.Dalhed, J.M.Green *et al.*, *J. Comput. Phys.* **32**, 212, (1979).
- [12] J.P.Goedbloed, *Comput. Phys. Commun.* **24**, 311 (1981).
- [13] J.Blum, J.L.Foil and B.Thooris, *Comput. Phys. Commun.* **24**, 235 (1981).
- [14] H.Lutjens, A.Bondesons and A.Roy, *Comput. Phys. Commun.* **69**, 287 (1992).

- [15] J.DeLucia, S.C.Jardin and A.M.Todd, J. Comput. Phys. **37**, 183, (1980).
- [16] K.M.Ling and S.C.Jardin, J. Comput. Phys. **58**, 300, (1985).
- [17] R.Gruber, R.Iacono and F.Troyon, J. Comput. Phys. **73**, 168, (1987).
- [18] G.T.A.Huysmans, J.P.Goedbloed and W.Kerner, in *Proceedings CP90 Conference on Computational Physics Proceedings* (Word Scientific, Singapore, 1991), p. 371.
- [19] B.Turkington, A.Lifschitz, A.Eydeland and J.Spruck, J. Comput. Phys. **106**, 269 (1993).
- [20] G.O.Ludwig, Plasma Phys. Controlled Fusion **39**, 2021 (1997).
- [21] R.Albanese and F.Villone, Nucl. Fusion **38**, 723 (1998).
- [22] L.L.Lao, H.E.St.John, R.D.Stambaugh, D.Stambaugh, A.G.Kellman and W.Pfeiffer, Nucl. Fusion **25**, 1611 (1985).
- [23] J.Blum, E.Lazzaro, J.O'Rourke, B.Keegan and Y.Stephan, Nucl. Fusion **30**, 1475 (1990).
- [24] J.R.Ferron, M.L.Walker, L.L.Lao, H.E.St.John, D.A.Humphreys and J.A.Leuer, Nucl. Fusion **38**, 1055 (1997).
- [25] L.LoDestro and L.D.Pearlstein, Phys. Plasmas **1**, 90 (1994).
- [26] J.F.Clark and D.J.Sigmar, Phys. Rev. Lett. **38**, 70 (1977).
- [27] S.P.Hirshman and S.C.Jardin, Phys. Fluids **22**, 731, (1979).
- [28] V.D.Khait, Sov. J. Plasma Phys. **6**, 476 (1980).
- [29] L.L.Lao, S.P.Hirshman and R.M.Wieland, Phys. Fluids **24**, 1431 (1981).
- [30] L.L.Lao, Comput. Phys. Commun. **31**, 201 (1984).
- [31] P.N.Vabishevich, L.M.Degtyarev and A.P.Favorski, Sov. J. Plasma Phys. **4**, 554 (1978).
- [32] L.M.Degtyarev and V.V.Drozdov, Comput. Phys. Rep. **2**, 341 (1985).
- [33] L.M.Degtyarev and V.V.Drozdov, Int. J. Mod. Phys. C **2**, 30 (1991).
- [34] S.Yu.Medvedev, L.Villard, L.M.Degtyarev, A.A.Martynov, R.Gruber and F.Troyon, 20th EPS Conf. on Controlled Fusion and Plasma Phys., Lisbon, Proc. Contrib. Papers, Vol. 17C, Part IV, 1279 (1993).
- [35] V.V.Drozdov, The POLAR-2D-NW code, Private communications.
- [36] R.R.Khayrutdinov, J.B.Lister, V.E.Lukash and J.P.Wainwright, Plasma Phys. Control. Fusion, **43**, 321 (2001).
- [37] J.-Y.Favez, R.R.Khayrutdinov, J.B.Lister, and V.E.Lukash, Plasma Phys. Control. Fusion, **44**,

171 (2002).

- [38] ITER Final Design Report. Control System Design and Assessment, Section 5.1.1 "Plasma Current, Position and Shape Control", G45 FDR 1 10-70-13 R1.0, 2001.
- [39] L.E.Zakharov and A.Pletzer, Phys. of Plasmas **6**, No. 12, 4693 (1999).
- [40] Ref. on the EFIT code
- [41] IAEA-CN-94/CT/P-08, ITER Preprints, INTERNATIONAL ATOMIC ENERGY AGENCY NINETEENTH FUSION ENERGY CONFERENCE, Lyon, France, 14-19 October 2002.
- [42] Samarsky A.A., Tishkin V.F., Favorsky A.P., et al. \ Sov. J. Differential Equations **7**, 1171 (1981).
- [43] L.M.Degtyarev and V.V.Drozдов, S.Yu.Medvedev. Equilibrium and stability numerical modeling of toroidal plasma, USSR Academy Science Book, M.V.Keldysh Institute of Applied Mathematics, Moscow 1989.